

# Arunabha Majumder

**R&D Engineer | Mechanical Engineering | New Product Development | 3D CAD (SolidWorks) | Testing & Validation**

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## PROFILE

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R&D Engineer with a degree in Mechanical Engineering, an MTech in Mechatronics and a PhD in progress, with over five years of research and development experience turning ideas into practical engineering solutions. Covers the full development cycle from concept generation and design through prototype fabrication, assembly, testing and validation. Produces detailed 3D CAD models, drawings and engineering documentation in SolidWorks, uses FEA to optimise for strength, stiffness and weight, and designs with manufacturability and cost in mind. Analytical problem solver with a proactive, innovative mindset and strong communication skills built across multidisciplinary international teams.

## KEY SKILLS

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- **R&D and Product Development:** Research and development of innovative product solutions, concept generation, new product development, prototype development, product improvement
- **3D CAD and Documentation:** Detailed 3D CAD models, engineering drawings and documentation in SolidWorks (CSWA certified) and Fusion 360
- **Testing and Validation:** Engineering testing methodologies, product testing, analysis and validation, custom test rigs, Finite Element Analysis (FEA)
- **Manufacturing Processes:** Fabrication, assembly, machined parts, additive manufacturing, Design for Assembly (DFA), design for manufacture
- **Optimisation:** Weight reduction, cost optimisation, part-count reduction, investigating emerging technologies
- **Professional:** Analytical and problem-solving skills, cross-functional collaboration, technical writing, project planning and delivery

## KEY ACHIEVEMENTS

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- Reduced energy consumption by more than 25% by developing a new hybrid actuation concept for a wearable device; awarded Best Research Paper, IFToMM ISRM 2026.
- Reduced product cost, mass and part count by eliminating an external force/torque sensor through a sensorless force-estimation method.
- Validated analytical design models against physical testing to below 10% RMS error, de-risking designs before fabrication.
- Secured an INR 100,000 TATA Technologies Innovation Grant and the Best Major Project Award for a rehabilitation product.

## EMPLOYMENT HISTORY

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**Research Engineer – R&D and Product Development (PhD Project: VIEXO) | 2023 – Present**  
*Aalborg University, Aalborg, Denmark*

- Led the design and development of an electromechanical product from concept generation through 3D CAD design, prototyping, testing and validation.
- Produced detailed SolidWorks 3D models, drawings and engineering documentation, and iterated designs on FEA results for strength, stiffness and weight before committing to fabrication.
- Fabricated and assembled prototypes from machined and additively manufactured parts, and integrated motors, encoders, force sensors and data acquisition.
- Designed custom automated test rigs and planned and delivered a 12-participant validation test programme across 3 operating modes and 2 load conditions, including test protocol, instrumentation and data analysis.
- Identified product improvements that cut energy consumption by more than 25% and reduced cost, mass and integration complexity by removing an external sensor.

- Collaborated across mechanical, electronics, control and biomechanics disciplines, and taught and supervised Bachelor's (B.Tech) students (2023 – 2025).

#### **MTech Researcher – Actuator Systems (Master's Thesis) | 2020 – 2022**

*CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India*

- Researched and developed a pneumatic actuator system: fabricated actuators in-house, built the test set-up with valves, pressure sensors and data acquisition, and characterised performance.
- Improved dynamic positioning accuracy to 0.3–0.78% tracking error with a neural-network-based controller, outperforming the conventional baseline under varying load.

#### **Mechanical Engineering Project – Upper-Limb Rehabilitation Device | 2018 – 2019**

*Siddaganga Institute of Technology, Tumakuru, India*

- Led CAD design, fabrication and user testing of a rehabilitation device from concept to working product.

### **EDUCATION**

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#### **PhD Candidate, Mechatronics and Robotics | 2023 – Present**

*Aalborg University, Department of Materials and Production, Denmark*

#### **MTech, Mechatronics | 2020 – 2022**

*AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), India*

#### **BTech, Mechanical Engineering | 2015 – 2019**

*Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), India*

### **CERTIFICATIONS AND AWARDS**

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- Certified SolidWorks Associate (CSWA) – Mechanical Design
- Best Research Paper Award, 9th IFToMM International Symposium on Robotics and Mechatronics (2026)
- TATA Technologies Innovation Award – INR 100,000 Grant (2019); Best Major Project Award (2019)

### **ENGINEERING PROJECTS**

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- **Go-Kart (2017):** Designed, fabricated and assembled a custom Go-Kart from the ground up.
- **Efficycle (2018):** Designed and built a human-electric hybrid vehicle in a multidisciplinary team for the Efficycle competition.

### **PUBLICATIONS**

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- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, MMS, vol. 213, Springer. (Best Research Paper Award)
- Majumder, A., et al. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." IEEE CONECCT, 2022.

### **TECHNICAL SKILLS**

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- **CAD and Simulation:** SolidWorks, Fusion 360, FEA
- **Software:** MATLAB/Simulink, Python, LabVIEW (basic), Arduino
- **Prototyping and Test:** 3D printing (Bambu Lab, Ultimaker Cura), force/torque sensors, motion capture, custom test rigs

### **LANGUAGES**

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English (C1, fluent) | Bengali (Native) | Hindi (Conversational)

### **REFERENCES**

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Available upon request.